WEEK 2 LAB TASKS - EXPLORATORY DATA ANALYSIS

Problem Context

You are provided with **three heterogeneous datasets** exhibiting **distinct data quality challenges**, including missing values, outliers, skewed distributions, and class imbalance.
Before applying any Artificial Intelligence or Machine Learning model, it is essential to design a **systematic, data-driven preprocessing pipeline** based on rigorous Exploratory Data Analysis (EDA).

This task requires you to apply **computational reasoning**, **statistical analysis**, and **structured preprocessing strategies** to transform raw datasets into **model-ready representations**.

Problem Statement

Design and implement an **EDA-driven preprocessing and feature preparation pipeline** for the following datasets:

- Iris dataset (clean, multi-class)
- Titanic dataset (missing values, class imbalance)
- Tips dataset (outliers, skewed numerical distributions)

You must perform all tasks in a single Jupyter Notebook with proper headings and outputs.

Task 1: Structured Dataset Acquisition and Structural Verification

Objective

To systematically acquire heterogeneous datasets and perform an initial structural assessment to understand data composition, attribute types, and integrity constraints.

Instructions

Using Seaborn, load the **Iris**, **Titanic**, and **Tips** datasets. For **each dataset**, perform a structured inspection by:

- Examining sample records from the beginning and end of the dataset
- Determining dataset dimensionality
- Enumerating all attributes
- Inspecting data types, null-value presence, and memory footprint

Expected Outcome

A verified understanding of dataset structure, attribute heterogeneity, and target variable identification.

Task 2: Attribute Role Identification and Data Typology Classification

Objective

To formally classify dataset attributes according to their computational role within an analytical pipeline.

Instructions

For **each dataset**, construct a classification table that assigns every attribute to one of the following roles:

- Numerical Feature
- Categorical Feature
- Target Variable

This classification must be exhaustive and mutually exclusive.

Task 3: Missing Data Quantification and Deterministic Imputation Strategy (Titanic Dataset)

Objective

To quantify missing data and apply a deterministic preprocessing strategy to restore dataset usability.

Instructions

- Compute column-wise missing value statistics.
- Apply the following **imputation and elimination rules**:

Attribute	**Preprocessing Action**
age	Mean-based imputation
embarked	Mode-based imputation
deck	Complete attribute removal

- Re-evaluate the dataset to ensure missing data elimination.

Note:

- **Mean imputation** for numerical attributes
- **Mode imputation** for categorical attributes

Task 4: Redundancy Detection and Duplicate Record Elimination

Objective

To enhance data integrity by identifying and removing redundant observations.

Instructions

For **each dataset**, detect duplicate records, eliminate them, and document the change in dataset dimensionality before and after removal.

Task 5: Descriptive Statistical Characterization of Features

Objective

To computationally summarize numerical features and analyze group-level statistical behavior.

Instructions

- **Iris Dataset**:
  Compute species-wise **mean** and **standard deviation** for all numerical attributes.
- **Titanic Dataset**:
  Compute survival-wise **mean** values for **age** and **fare**.
- **Tips Dataset**:
  Compute **mean** and **median** values for **total_bill** and **tip**.

Task 6: Distributional Analysis through Univariate Visualization

Objective

To analyze feature distributions and variability through graphical representations.

Instructions

Generate **histograms** for the following attributes:

- **Iris:** sepal_length, petal_length
- **Titanic:** age, fare
- **Tips:** total_bill, tip

Each visualization must be properly labeled and formatted for analytical clarity.

Task 7: Statistical Outlier Identification Using Interquartile Range (IQR)

Objective

To algorithmically identify anomalous observations based on robust statistical boundaries.

Instructions

Using the **Tips dataset**, compute Q1, Q3, and IQR for **total_bill** and **tip**.
Determine **lower** and **upper** bounds and identify observations falling outside these limits.

Generate **box plots** for:

- total_bill
- tip

Calculate bounds using:

Count number of outliers per column

No removal in this task

Task 8: Outlier Mitigation via IQR-Based Row Elimination

Objective

To refine dataset quality by removing statistically identified outliers.

Instructions

Eliminate all records violating IQR bounds (as computed in Task 7) and generate a cleaned dataset.
Document dataset size before and after outlier removal.

Task 9: Post-Processing Statistical Reassessment

Objective

To assess the computational impact of outlier mitigation on data characteristics.

Instructions

Recompute **mean**, **median**, and **standard deviation** for affected attributes **before and after** outlier removal and present results in a comparative format.

Task 10: Bivariate Relationship Analysis via Scatter Plots

Objective

To examine inter-feature relationships and potential correlations.

Instructions

Generate **scatter plots** for:

- **Iris:** sepal_length vs petal_length
- **Titanic:** age vs fare
- **Tips:** total_bill vs tip

Comment on:

- Linearity
- Clustering
- Strength of relationship

Task 11: Class Imbalance Detection and Corrective Sampling Strategy (Titanic Dataset)

Objective

To identify and mitigate **class imbalance** that may adversely affect predictive modeling.

Instructions

- Compute the frequency distribution of the target variable **survived**. (Using the preprocessed Titanic dataset (after missing value handling and duplicate removal)
- Visualize class proportions using a **count plot**

Mandatory Imbalance Handling

Apply **random undersampling** to equalize class representation.

- Reduce the majority class (survived = 0)
- Match its size to the minority class (survived = 1)

Store the balanced dataset in a **new DataFrame**.

Mandatory Verification

Validate class balance post-processing through numerical and visual confirmation.

Recompute:

- Class counts after balancing
- Dataset shape before and after balancing

*Plot class distribution **after balancing**.*

Visualize class distribution using:

- Bar chart

Task 12: Correlation Analysis and Feature Redundancy Identification

Objective

To identify statistically redundant features using correlation-based analysis.

Instructions

Compute **correlation matrices** for all datasets.

Identify:

- Strong positive correlations
- Strong negative correlations
- **One highly correlated feature pair per dataset**, indicating potential dimensional redundancy.

Task 13: Structured Analytical Summary Report

Objective

To consolidate EDA findings into a structured analytical summary.

Instructions

Prepare **exactly ten bullet points** summarizing:

- Data quality issues
- Preprocessing operations performed
- Impact of transformations
- Dataset readiness for AI/ML modeling

Submission Guidelines:

- Download .ipynb from Jupyter
- Push it to your GitHub Repository named **yourRollno_Notebooks**
- Upload only link through attachment option on GCR